



LESSON - 6



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INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

CONSTRAINTS

SQL constraints are used to specify rules for the data in a table.

Want clean and optimized databases? Use SQL Constraints! ☒ NOT NULL stops empty values, UNIQUE prevents duplicates, PRIMARY KEY uniquely identifies records, and FOREIGN KEY links tables. Want faster searches? Use INDEX!



Type of Constraints

The following constraints are commonly used in SQL:

NOT NULL - Ensures that a column cannot have a NULL value;

UNIQUE - Ensures that all values in a column are different;

PRIMARY KEY - A combination of a NOT NULL and UNIQUE. Uniquely identifies each row in a table;

FOREIGN KEY - Prevents actions that would destroy links between tables;

CHECK - Ensures that the values in a column satisfies a specific condition;

DEFAULT - Sets a default value for a column if no value is specified;

CREATE INDEX - Used to create and retrieve data from the database very quickly;

All in One

The following constraints are Students Table used in SQL:

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,    -- Unique identifier (NOT NULL + UNIQUE)  
    name VARCHAR(50) NOT NULL,    -- Name cannot be NULL  
    email VARCHAR(100) UNIQUE,    -- Ensures unique email addresses  
    age INT CHECK (age > 5),      -- Age must be greater than 5  
    grade VARCHAR(20) DEFAULT 'Not Assigned', -- Default grade  
    class_id INT,                 -- Foreign key reference  
    FOREIGN KEY (class_id) REFERENCES Classes(class_id) -- Ensures class exists  
);  
  
-- Create INDEX for faster name searches  
CREATE INDEX idx_student_name ON Students(name);
```

NOT NULL

Ensures that a column cannot have a NULL value..

```
CREATE TABLE Employees (  
    EmployeeID INT NOT NULL,  
    FirstName VARCHAR(50) NOT NULL,  
    LastName VARCHAR(50)  
);
```

UNIQUE

Ensures all values in a column are different.

```
CREATE TABLE Customers (  
    CustomerID INT UNIQUE,  
    Email VARCHAR(100) UNIQUE  
);
```

CHECK

Ensures that the values in a column satisfy a specific condition.

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY,  
    Price DECIMAL(10, 2) CHECK (Price > 0),  
    Stock INT CHECK (Stock >= 0)  
);
```

DEFAULT

Sets a default value for a column if no value is specified.

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    OrderDate DATE DEFAULT (CURRENT_DATE),  
    Status VARCHAR(20) DEFAULT 'Pending'  
);
```


CREATE INDEX

Used to create and retrieve data from the database very quickly.

```
CREATE TABLE Sales (  
    SaleID INT PRIMARY KEY,  
    ProductName VARCHAR(50),  
    SaleDate DATE  
);  
  
-- Creating an index on the SaleDate column  
CREATE INDEX idx_SaleDate ON Sales(SaleDate);
```


Auto Increment

Auto-increment allows a unique number to be generated automatically when a new record is inserted into a table.

```
CREATE TABLE Persons (  
    Personid int NOT NULL AUTO_INCREMENT,  
    LastName varchar(255) NOT NULL,  
    FirstName varchar(255),  
    Age int,  
    PRIMARY KEY (Personid)  
);
```